

Wages and Employment in a Single Labour Market

ECON 300 - Labour Economics

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Learning Objectives

- Construct market supply and demand and compute equilibrium wage and employment.
- Contrast competitive outcomes with imperfect competition (monopoly, oligopoly, monopolistic competition).
- Explain how monopsony reduces wages and employment relative to competition.

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- How does imperfect competition modify the supply-demand analysis?
- Who bears payroll taxes – workers or firms?
- What is monopsony? How different are outcomes when firms have hiring power?

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The Competitive Firm's Interaction with the Market

- Competitive in both product and labour markets
- Homogeneous labour; firm is a price and wage taker
- Firm can hire all needed labour at the market wage
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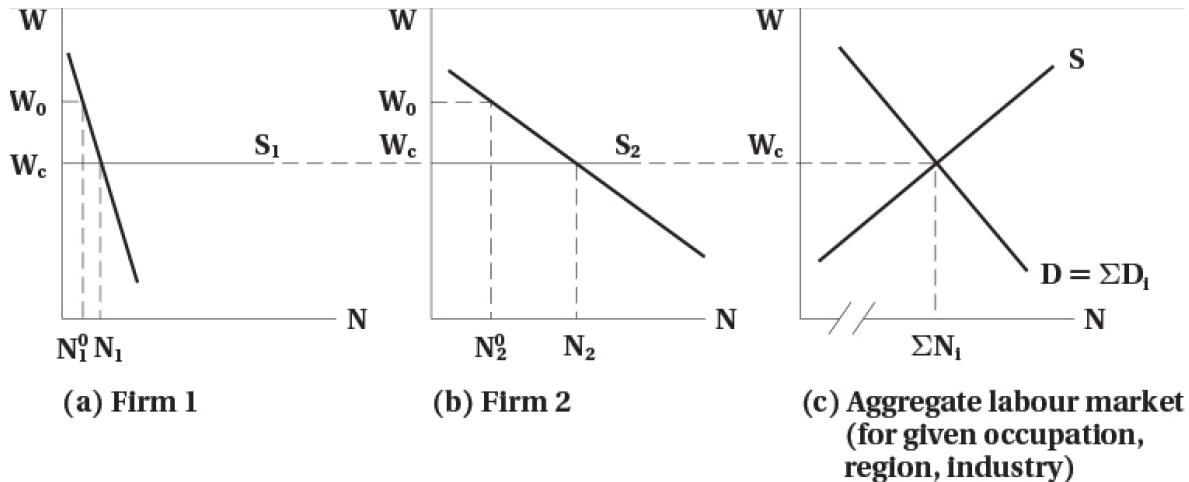
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Figure 7.1: Competitive Product and Labour Markets



Panels (a) and (b): Individual Firm Labour Demand

Firm Labour Demand

- Downward-sloping curve = **value of marginal product of labour (VMPL)**
- Hiring rule:

$$\text{Hire labour until } VMPL = W$$

- Each firm faces a **perfectly elastic labour supply** at the market wage

Effect of a Wage Change

- At wage W_0 , Firm i hires N_i^0
- At lower wage W_c , labour hired increases to N_i
- Firms hire more labour when wages fall

Key Insight

- Individual firms take wages as given
- Labour demand responds to wage changes

Panel (c): Aggregate Labour Market

Market Labour Demand

- Market demand is the **horizontal sum** of firm demands:

$$D = \sum_i D_i$$

- At wage W_0 :

$$N^{market} = N_1^0 + N_2^0 + \dots$$

Labour Supply

- Upward-sloping curve S represents market labour supply

Equilibrium

- Wage W_c clears the market
- Labour supply equals labour demand
- Firms hire where $W_c = VMPL$

Labour Supply in Figure 7.1

Labour Supply at the Firm Level (Panels a & b)

- Each firm faces a **perfectly elastic labour supply** at the market wage
- Represented by horizontal lines (S_1 , S_2)
- Reason:
 - Individual firms are wage takers
 - They can hire as many workers as desired at the prevailing wage
 - Hiring decisions do *not* affect the wage

Labour Supply in the Market (Panel c)

- Market labour supply curve (S) is **upward sloping**
- Reflects workers' willingness to supply more labour at higher wages
- Determined by:
 - worker preferences (income vs leisure)
 - alternative employment opportunities
 - demographics and participation decisions

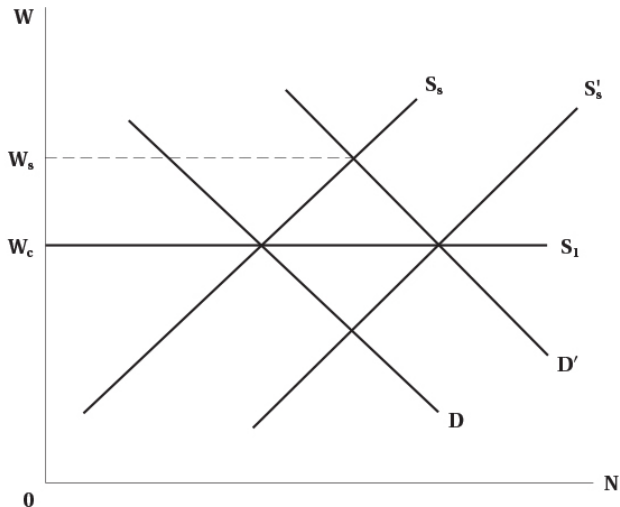
Short Run vs. Long Run to the Firm

- Short run: Product demand rise $\Rightarrow$ higher labour demand; wage and employment rise (given upward-sloping supply to the firm).
- Long run: Higher short-run wages attract more workers; labour supply shifts right; wage adjusts.

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Figure 7.2: Short-Run and Long-Run Labour Supply Schedules



Short-run:

- Even a competitive firm may have to raise its wages to attract additional workers
 - dynamic monopsony
- The firm's short-run labour supply schedule could be upward sloping, S_s

Long-run:

- Since $W_s > W_c \Rightarrow$ there will be supply influx of additional workers
- Resulting in the shift of supply schedule from S_s to S'_s
- The supply influx would depress the temporarily high, short-run wage of W_s back to its long-run level of W_c

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Monopoly: Effects of Hiring More Labour

- $MPP_N \downarrow$ with more N
- MR falls as more output sold requires lower price
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Figure 7.3: Monopolist vs. Competitive Demand for Labour

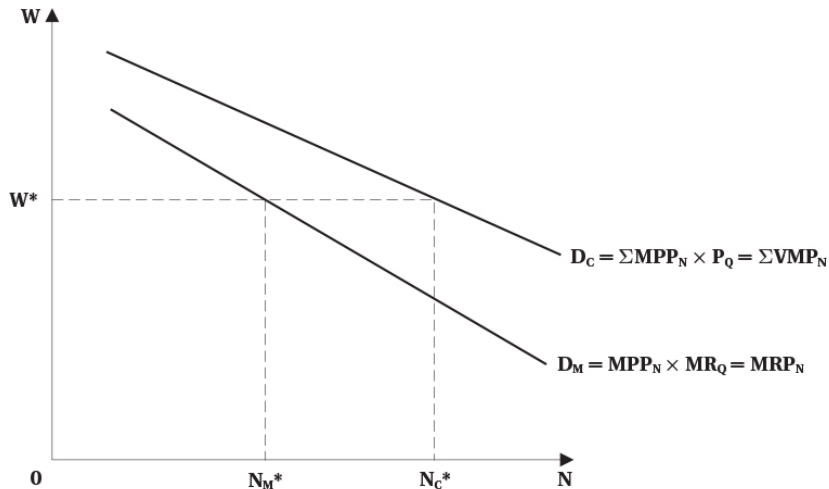


Figure 7.3: Monopolist vs Competitive Demand for Labour

Competitive Labour Demand (D_C)

- Firms are price takers in the product market
- Output price is constant: P_Q
- Labour demand equals the **value of marginal product**:

$$D_C = VMP_N = MPP_N \times P_Q$$

- Higher employment level: N_C^*

Monopolist Labour Demand (D_M)

- Marginal revenue is less than price: $MR_Q < P_Q$
- Labour demand equals the **marginal revenue product**:

$$D_M = MRP_N = MPP_N \times MR_Q$$

- Curve lies below D_C and is steeper – MR_Q falls as employment and output increase
- Lower employment level: N_M^*

Oligopoly & Monopolistic Competition

- Oligopoly: few firms; strategic interdependence
- Possible departures from market wages (above-normal profits may be partly shared)
- Monopolistic competition: short-run profits $\Rightarrow$ possibly higher-than-market wages; long-run zero profit $\Rightarrow$ market wages

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Linear Supply and Demand: Setup

$$N_D = a + bW; b < 0, \quad N_S = c + fW; f > 0$$

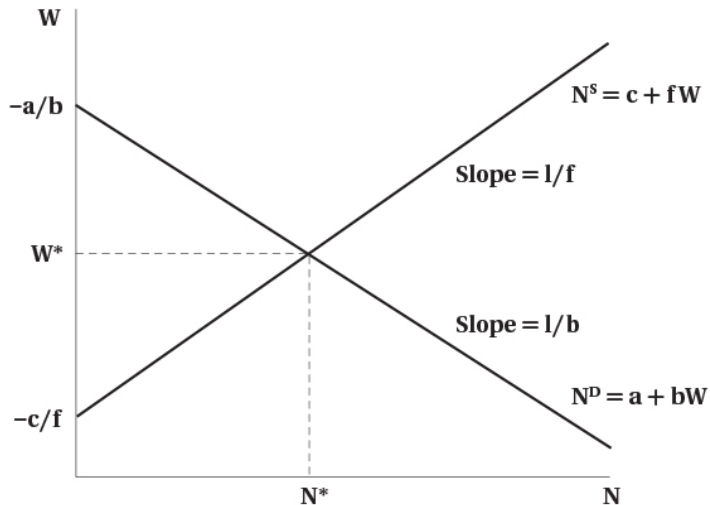
Setting $N_S = N_D$ yields equilibrium (w^*, N^*) .

With Shifter z :

$$N_D = a + bw + \gamma z, \quad N_S = c - dw + \delta z$$

Solve again for new (w^*, N^*) to study incidence and comparative statics.

Figure 7.4: Linear Supply and Demand



Application: Incidence of a Unit Payroll Tax

- Examples: CPP/QPP, workers' compensation, EI, health/education payroll levies
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TABLE 7.1: Payroll Taxes as a Percentage of Labour Income, Canada (1971, 1997)

Type of Payroll Tax	1971	1997
Workers' compensation	0.73	1.61
Unemployment insurance	1.06	4.98
Canada/Quebec pension	2.12	3.92
Health and/or education	0.21	1.71
Total	4.12	12.23

Payroll Tax in the Model

Let a per-worker tax t on firms shift the perceived cost:

$$N_D = c - d(w + t)$$

Incidence on wages vs. employment depends on the elasticities of N_S and N_D .

Figure 7.5: Effect of a Payroll Tax on Employment and Wages

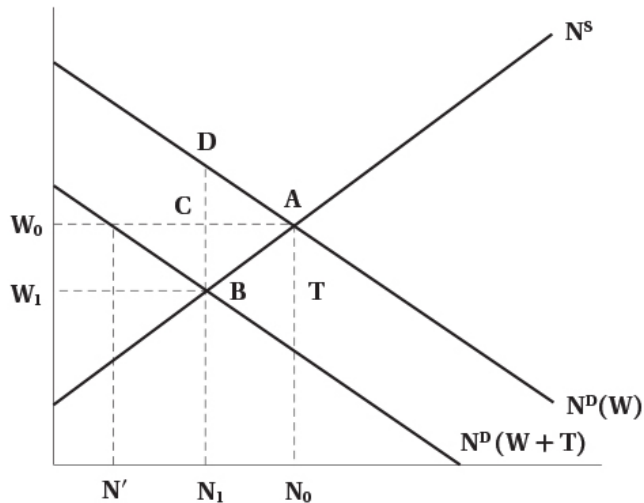


Figure 7.5: Payroll Tax, Wages, and Employment

Initial Equilibrium

- Labour supply: N^S
- Labour demand: $N^D(W)$
- Initial equilibrium at point A
- Wage = W_0 , Employment = N_0

Introduction of a Payroll Tax (T per worker)

- Tax raises cost of labour to employers to $W + T$
- Labour demand shifts downward to $N^D(W + T)$
- New equilibrium at point B
- Workers receive lower wage: W_1
- Employment falls from N_0 to N_1

Figure 7.5: Payroll Tax, Wages, and Employment

Tax Incidence

- Total tax per worker = T
- Workers bear part of the burden:

$$\text{Worker share} = BC$$

- Employers bear the remainder:

$$\text{Employer share} = CD$$

Summary

- Payroll taxes reduce employment and wages
- Tax burden is shared between workers and firms
- The side of the market that is less elastic bears more of the tax burden

Monopsony: Basics

- A monopsony is a labour market with one dominant firm.
- Large firm relative to labour market size; influences wage
- It faces an upward-sloping labour supply curve
- To hire more workers, the firm must offer higher wages.
- Marginal cost of labour exceeds average cost (wage) due to **inframarginal uprating**

Inframarginal uprating is an increase in benefit levels for individuals already eligible for a program.

- Eligibility rules remain unchanged.
- Raises payments to current recipients
- Does not expand eligibility
- Does not directly add new participants

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Example

- Unemployment benefits increase from \$500 to \$550 per week for current recipients.

Economic Effects

- Increases recipients' income
- May reduce labour supply through an income effect
- Participation incentives at eligibility thresholds remain unchanged

Contrast

- **Marginal reform:** A policy that expands eligibility or lowers the cutoff – affects marginal entrants, not inframarginal recipients.

Figure 7.6: Monopsony

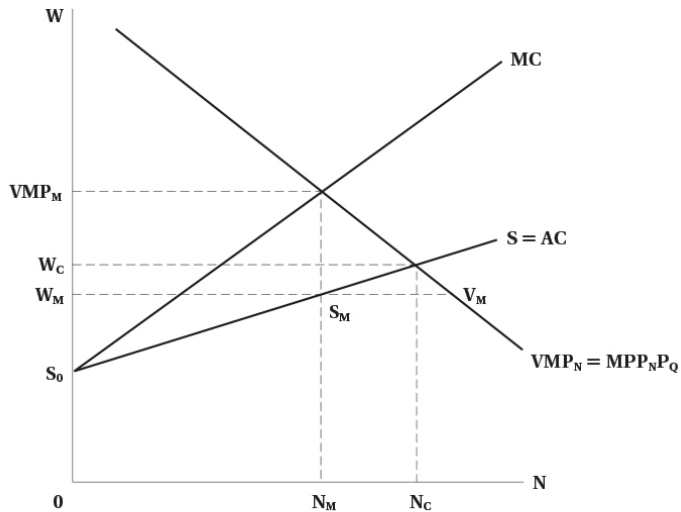


Figure 7.6: Monopsony in the Labour Market

Curves in the Figure

- Labour supply ($S = AC$): wage required to attract workers; also the average cost of labour.
- Marginal cost of labour (MC): lies above supply since hiring one more worker requires raising wages for all workers.
- Labour demand (VMP_N):

$$VMP_N = MPP_N \times P_Q$$

downward sloping due to diminishing productivity.

Monopsony Outcome

- Profit maximization occurs where $MC = VMP_N$.
- Employment: N_M
- Wage paid: W_M (from the supply curve)

Figure 7.6: Monopsony in the Labour Market

Competitive Benchmark

- Competitive equilibrium: N_C and W_C
- Monopsony results:

$$N_M < N_C \quad \text{and} \quad W_M < W_C$$

Key Point

- Market power allows the firm to restrict employment and suppress wages.

Perfect Monopsonistic Wage Differentiation

Definition

- Occurs when a monopsonist pays each worker a wage equal to their **reservation wage**.
- The firm has full information about workers' minimum willingness to work.
- Workers receive different wages instead of a single uniform wage.

Key Mechanism

- Hiring an additional worker does *not* require raising wages for existing workers.
- The labour supply curve becomes the firm's marginal cost of labour:

$$MC = \text{Labour Supply}$$

Hiring Rule

- The firm hires labour where:

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Employment Outcome

- Employment equals the competitive level.
- The efficient number of workers is hired.

Distribution of Surplus

- Workers receive wages equal to their reservation wages.
- Worker economic rent (the surplus income a worker receives above their reservation wage) is eliminated.
- The firm captures all surplus.

Comparison

- Competitive market: single wage, surplus shared.
- Single-wage monopsony: lower wage, reduced employment.
- Perfect wage discrimination: efficient employment, firm captures surplus.

Real-World Relevance

- Pure wage discrimination is rare.
- Partial forms appear through:
 - individualized contracts
 - performance pay
 - bargaining and bonuses
 - non-wage compensation

- Competitive equilibrium determines wage and employment; product market power lowers labour demand.
- Payroll tax incidence depends on elasticities; wages and employment adjust accordingly.
- Monopsony lowers wage and employment; targeted minimum wage can raise both.
- Measuring minimum wage effects empirically remains contested.

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